



INTE130 – Object-Oriented Programming

Designing a One-Person Company Using a Multi-Agent System .

Submitted Parts: Part A & Part B

Prepared by :

Sultan Khalid Al Yarabi - 2318592

Malik Marzooq Al Rahbi - 2527871

Khamis Hamed Alowaisi - 2422670

Mohammed Naeem Al Gheilani – 2526168

Instructor Name:

Dr. Akbar khanan

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Part A – Understanding One-Person Company Systems

1. One-Person Company Concept

Definition

A One-Person Company (OPC) is a legal business structure that allows a single individual to own, manage, and control a company independently. It provides limited liability protection, meaning the owner's personal assets are protected beyond the invested capital.

Who does it serve?

- Individual entrepreneurs and startups
- Freelancers and self-employed professionals
- Small business owners seeking formal company registration
- People testing business ideas with lower risk

How does it work?

- One shareholder owns and controls the company
- A nominee is appointed in case of death/incapacity
- Must follow legal requirements (registration, tax, accounting)
- Limited liability separates personal and business assets
- No board of directors → faster decision-making

Role of Digital Systems

- Accounting and financial management software
- Digital marketing (social media, email campaigns)
- Data storage and business analytics
- Automation of tasks (orders, scheduling, communication)
- Remote operation without physical office

Real-world Examples

- India: OPC model introduced under Companies Act 2013
- Freelancers (developers, designers) operating single-owner companies
- E-commerce sellers on Amazon/Shopify using digital systems

2. Architecture of a One-Person Company

One-person company can be characterized as:

- **Founder / Owner:**
He is the business owner who manning all process and have the full authority on the company.
- **Digital Agents:**

Programs and systems helping in doing tasks instead of staff, and it is supporting the daily operations.
- **Business Tasks:**
It is group of tasks done for business-like financial managing, marketing, customer services, and processes managing.
- **System Components:**
System contains several things that assess it to work like registration system and data managing, communication tools, accountant systems, and business operating platform.

3. Role of Multi-Agent Systems

Multi-Agent system helps the owner tin distributing tasks widely as every agent is responsible for a specific task so we can start multi tasks at one time, so that the load will be reduced and it increases efficiency and speed of operation.

So, the owner can do all tasks alone.

4. Programming Perspective

OPC can be represented using OOP as follows:

- **Class Founder:**
It acts for the company owner and contains the authority of decision making.
- **Class Agents:**
Digital agents who are taking tasks, as any of them have his own function.
- **Class Company:**
Representing the company and it holds the data and process.
- **Class Tasks:**
Represents the various tasks within the company.

Part B – Business Idea and Project Proposal.

1- Business Name: CyberShield system

2- Business Background:

CyberShield is a one-person company that provides cybersecurity services to small and medium-sized businesses. Its primary goal is to protect corporate systems, networks, and data from cyber threats. The system helps monitor activity, identify vulnerabilities, and offer suggestions for improving security. Many companies cannot afford a full cybersecurity team or staff, so CyberShield provides a simple and effective solution using digital agents managed by a single founder.

3- Target Customers:

The target customers are small and medium-sized enterprises (SMEs), e-commerce companies, and startups. These companies or entities naturally need protection for their systems, but as mentioned, they often lack the budget or sufficient staff to hire a

cybersecurity team. Therefore, they can utilize this system, which is managed by a single individual with the assistance of skilled agents.

4- Why the Business can Be Managed by One Person:

This work can be managed by one person because most tasks are performed by digital agents. The founder makes the decisions, while these agents carry out the required tasks such as threat monitoring, vulnerability scanning, and reporting. This makes the work easier and faster, and allows one person to manage the entire system.

5- Proposed Agents:

1) Financial Agent

How it works:

- **Collects and analyzes operational cost data to support financial decision-making.**
- **Tracks spending, token usage, and resource consumption in real time.**
- **Estimates business impact and prioritizes actions based on financial risk.**
- **Produces executive-level summaries to support budgeting and planning.**

Interaction with other agents:

- **Receives incident-cost information from the Incident Response Agent.**

- **Uses risk findings from the Vulnerability Analysis Agent to estimate remediation costs.**
- **Receives operational metrics from the Reporting Agent for financial analysis.**

2) Threat Monitoring Agent

How it works:

- **Continuously monitors logs, endpoints, and network events in real time.**
- **Detects anomalies such as suspicious login attempts or abnormal behavior patterns.**
- **Correlates threat intelligence signals to identify possible attacks earlier.**
- **Generates alerts automatically when malicious activity is detected.**

Interaction with other agents:

- **Sends alerts to the Incident Response Agent for immediate action.**
- **Shares suspicious behavior data with the Vulnerability Analysis Agent to identify root causes.**
- **Forwards monitoring results to the Reporting Agent for dashboards and reports.**

3) Vulnerability Analysis Agent

How it works:

- **Continuously scans systems to discover exposed assets and weaknesses.**
- **Detects weak configurations, outdated software, and exploitable vulnerabilities.**
- **Prioritizes vulnerabilities by severity and business impact.**
- **Recommends remediation actions such as patching or configuration changes.**

Interaction with other agents:

- **Sends discovered weaknesses to the Incident Response Agent for remediation.**
- **Provides high-risk vulnerability data to the Threat Monitoring Agent for focused monitoring.**
- **Sends findings to the Reporting Agent for executive summaries.**

4) Incident Response Agent

How it works:

- **Receives alerts from detection systems and classifies incident severity.**
- **Executes containment, eradication, and recovery actions automatically or semi-automatically.**
- **Applies response playbooks such as blocking access or isolating systems.**

- **Records response actions for audit and future improvement.**

Interaction with other agents:

- **Receives alerts from the Threat Monitoring Agent.**
- **Uses data from the Vulnerability Analysis Agent to understand exploited weaknesses.**
- **Sends incident results and response logs to the Reporting Agent.**
- **Shares cost-impact data with the Financial Agent.**

5) Reporting Agent

How it works:

- **Aggregates findings from monitoring, vulnerability, and incident-response systems.**
- **Produces executive and technical reports automatically.**
- **Includes detected threats, vulnerabilities, actions taken, and recommendations.**
- **Generates compliance-ready reporting for management review.**

Interaction with other agents:

- **Receives data from all other agents.**
- **Combines technical and financial insights into unified reports.**
- **Provides feedback loops to improve monitoring and response decisions.**

6- Example Interaction

The founder assigns a task such as "system security check." The monitoring agent detects suspicious activity, and then the vulnerability agent examines the system. Next, the response agent suggests appropriate solutions, and finally, the reporting agent prepares a report. Based on this, the founder makes the final decision.